SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
PRACTICAL ASSIGNMENT 12

1. Student Details

Student Name	Homesh Arvind Sawarkar
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Division	A
Batch	A2
Roll No.	202501110069
Date of Submission	27/10/2026

2. Practical Details

Practical No.	12
Practical Title	TYPE CASTING
Date Performed	-
Faculty Name	Khushal Khairnar

3. Problem Statement

Assignment based on type casting and casting operators.

4. Objectives

1. To understand the concept of type casting and its importance in C++ programming.
2. To implement different types of type casting, including implicit and explicit type conversion.
3. To learn and apply C++ casting operators such as static_cast, dynamic_cast, const_cast, and reinterpret_cast.

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Explain the difference between implicit and explicit type casting in C++.
2. Use casting operators to convert data types and handle type conversions appropriately.
3. Implement different casting operators and understand their applications in C++ programs.

6. Theory / Concept

Type casting is the process of converting a value from one data type to another. It is useful when performing operations involving different data types or when a specific type conversion is required.

The main types of type casting in C++ are:

1. Implicit Type Casting: The compiler automatically converts one data type into another. For example, converting an integer to a floating-point number.
2. Explicit Type Casting: The programmer manually converts a value into another data type using a cast.

7. Algorithm / Pseudocode

START

Step 1: Include the required header files and declare variables of different data types.

Step 2: Demonstrate implicit type casting by assigning an integer value to a floating-point variable.

Step 3: Demonstrate explicit type casting using static_cast to convert a floating-point value into an integer.

Step 4: Create a polymorphic base class and derived class to demonstrate dynamic_cast. Also demonstrate const_cast and reinterpret_cast using appropriate variables and pointers.

Step 5: Display the results of the conversions and verify the behavior of each casting operator.

STOP

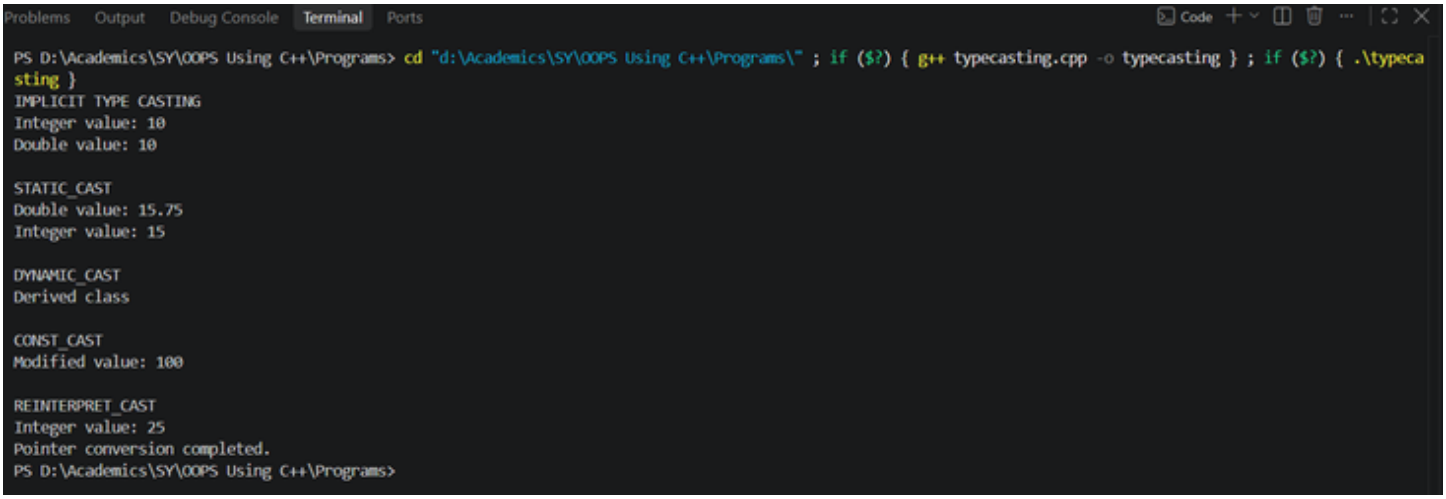
08. Technology / Tools / Libraries Used

Technology / Tool	**Purpose**
Programming Language	C++
IDE / Platform	VS Code
Libraries / Packages	-

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	https://github.com/Homeshsawarkar/Assignment-12-C-
Repository Name	Assignment 12 C++
Branch Used	main
Commit ID / Latest Commit	•
Date of Last Commit	Oct 27, 2026

10. Results / Observations



11. Conclusion

In this practical, type casting and casting operators in C++ were studied and implemented. The program demonstrated implicit conversion, explicit conversion using static_cast, runtime casting using dynamic_cast, qualifier conversion using const_cast, and low-level pointer conversion using reinterpret_cast. This practical helped develop an understanding of type conversion and the appropriate use of casting operators in C++.

12. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
Homesh Sawarkar	202501110069	Homesh A.S